F71AA Actuarial Applications: Business Writing

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5 Business Writing

Most of this course (and much of your degree!) will be mathematics or the use of software. Actuaries are highly regarded for these technical skills. The purpose of those skills is to help insurers, pension schemes and more to make the best possible decisions. For that, you must be able to communicate your findings effectively to others.

This is also why we often require you to include a short report as part of your assessments. Part of the purpose of this chapter is to help you understand what your lecturers (and the IFoA) think a good report looks like.

5.1 Introduction

It is important that as actuaries, we are able to communicate our work clearly with those that we are working with. In many cases, actuaries will be working with other people who will not have specialist knowledge of the technical content that actuaries work on. For example, actuaries often work closely with the sales team of an insurance company who are unlikely to be comfortable with complicated mathematics or actuarial terminology. It is therefore essential that actuaries can communicate their important messages easily so that they are well understood.

In this chapter, we will look at general principles for communication ("who? why? what?"), some general tips, requirements for specific types of communication (e.g., reports) and finally examples of good an bad communication.

The golden rule when choosing how you communicate is that your choices should all **help the reader** (or listener) **understand your message**. Therefore, you need to know your reader's needs (e.g., will they understand technical language?) and you need to be very clear about what message you want to get across (i.e., what do you want to achieve with your communication?)

5.2 Who?

It is important to keep in mind who you are communicating with. In order to maximise the chance of the recipient understanding the message easily, it is important that the communicator keeps the recipient in mind from the outset. This is particularly important when choosing the language and structure of the communication.

5.3 Why?

The objectives of the communication should be considered from the outset. The objectives might include:

- Explaining information.
- Outlining a process.
- Discussing advantages and disadvantages of various proposals.
- Changing a viewpoint.

- Providing reassurance.

5.4 What?

Some common forms of written communication that actuaries must be comfortable with preparing include:

- Emails
- Memos
- Reports
- Presentations
- Articles/ Technical papers/ Briefing papers
- Policy documents

5.5 General tips (all communication forms)

5.5.1 Headings

Use headings to help the reader. The help act as a signpost helping the reader to navigate through the information that is presented. They also show the structure of the document at a glance. Make sure that the headings are succinct and clear. They should stand out from the other text of the document e.g., using bold or underlined font.

5.5.2 Length

The mathematician Blaise Pascal once wrote *"I would have written a shorter letter, but did not have the time."* It can be harder to write a good short communication than a long one, but it is often worth the effort.

Many actuaries will tell you that there are few things their clients like less than reports that are too long. A good principle is that a report should be as long as it needs to be and no longer. Leaving out unnecessary material or moving less important material to an appendix is a valuable part of an actuary exercising judgment. Arguably, if you do not know what in your expert opinion should be important or unimportant to your client, then you have not yet done your job.

However, you must be careful to find the right balance. You should not leave out anything that might affect the user's decisions or which is required by professional standards. Sometimes repetition (e.g., in a conclusion or executive summary) can help the user and justify the additional length.

5.5.3 Ordering material

You can take a "bottom up" or "top down" approach to communication. A "bottom up" approach is like a novel. You start from the beginning (e.g., "I set up a model using these assumptions") and finish at the end (e.g., "This is what I conclude from the results of my model"). This is often the easiest approach for the writer but it is not always best for the reader.

A "top down" approach starts with your conclusions or headlines, then explains the reasoning and evidence. This is often better for the reader. In that way, the writer acts more like a tour guide than a novelist. You don't want to keep your readers in suspense.

In practice, most communications will be somewhere between the two extremes. For example, even a "bottom up" report will generally include an executive summary.

5.5.4 Layout

Paragraphs should be used to make the content of your communication more digestible. One subject or idea should be explored per paragraph. The opening sentence of each paragraph should introduce the idea or subject. The final sentence of each paragraph should either: summarise, lead into the next paragraph or emphasise the key idea.

Bullet points should be used when you have items to list. However, be careful not to overuse bullet points because they can lose their impact in these scenarios.

Space out your writing to make it easier to read.

5.5.5 Sentences, Language and Tone

Sentences should be short and concise. Long sentences are difficult to read or understand. Ensure there is one idea per sentence.

For example, read the sentence below:

"IFRS 17, the International Financial Reporting Standard for insurance contracts issued by the International Accounting Standards Board (IASB), is a comprehensive and complex accounting standard that aims to provide transparent and consistent reporting of insurance contracts, requiring extensive data collection, sophisticated actuarial modeling, and significant implementation efforts from insurance companies worldwide."

Now read the sentences below that convey the same message:

"IFRS 17 is an accounting standard developed by the International Accounting Standards Board (IASB) for insurance contracts. It aims to provide transparent and consistent reporting of insurance contracts. Implementing IFRS 17 requires extensive data collection, sophisticated actuarial modeling, and significant effort from insurance companies worldwide. "

Avoid waffle and make your points accurately and precisely.

Use a friendly, personal and polite tone. Avoid being too formal or trying to sound 'too clever'. Your readers will normally assume that you are clever, so you don't need to use sophisticated language to prove it. Remember that your role is to help the reader: Clear, simple language suitable for the reader is normally far more appropriate.

5.5.6 Jargon

You should always avoid jargon unless it will help the reader. Defining jargon terms is rarely a good solution, even though using them might be convenient for the writer.

If your reader is a fellow actuary, especially a colleague doing similar work, then using jargon may be appropriate because you will both understand it. However, jargon is not suitable for policyholders or pension scheme members. The choice is not necessarily driven by the sophistication of the reader. For example, if you write a report for your Chief Financial Officer, you should not necessarily expect her to know the same terminology that you use day-to-day.

5.5.7 Communicating numerical information

As an actuary, a large part of your job will be communicating numerical information. Numerical information can help the reader understand a concept or may be needed to support your arguments/message.

In any communication, you need to decide:

- What numerical information to include (sometimes it may be more appropriate to leave it out)?

- What format should the numerical information be displayed?

The trick to communicating numerical information effectively is to present just enough numerical information for the reader to understand your message but not so much that the numbers distract from the message.

For example, if you need to explain to a policyholder how their pension pot has changed over the year. You do not need to show how the pension pot moved by using monthly data (12 data points). You can instead achieve this by using 2 data points - the value at the start and the end of the year. Illustrating more data would be distracting and confusing.

Numerical examples can often help the reader understand the key message. However, it is important the numerical example is short and simple. Using a complex example with lots of figures can distract from the explanation that you are trying to support.

5.5.8 Visual aids

Visual aids are useful in ensuring that your communication has a strong impact and are a useful way to support your explanations.

Diagrams, graphs and tables can be particularly useful in displaying numerical information.

Graphs
Graphs are an efficient way of displaying a large volume of information. However, it is important to be careful not to display too much information that is not needed.

To help ensure that your graphs are clear: · ·

- Give each graph a title
- Label the axes of graphs fully
- Provide a legend for graphs with more than one data set.

Some graphs that can be useful include:

- Line graphs - useful for showing trends (e.g., life expectancy over the last twenty years).
- Bar charts - useful for making comparisons (e.g., comparing gross written premiums this year versus last year across different product lines).
- Pie charts - useful for showing the split of a total (e.g., split of assets in a fund)

Tables
If the volume of information is sufficiently small, it may be more straightforward to present the information in a table of figures instead of a graph. The table should be simple and concise.

To help ensure that your tables are clear:

- Label the rows and columns of your table
- Clearly label the units for the numerical information presented.

If the table of figures is really large, this would belong in an appendix rather than in the main body of the communication.

5.5.9 Referencing

You are expected to source your own reference material to help inform your written communication. This means that you must include a bibliography of all the references which inform your writing, including author/organisation, year, title and (where relevant) weblink and date that you accessed the weblink.

Referencing is needed to

- Build the reader's trust in what they are reading
- To say where we have used someone else's fact, idea or opinion
- To quote someone else's worsds

Here is an example to illustrate:

What is AI?

AI stands for Artificial Intelligence, which refers to the simulation of human intelligence in machines that are designed to think and act like humans. AI technology is used to develop systems that can perform tasks that typically require human intelligence, such as recognizing speech, making decisions, and solving problems.

AI is opening up new possibilities in the insurance sector (Balasubramanian et al 2021) and, in particular, the life insurance sector (Balasubramanian et al 2020). Maier et al (2020) discuss the use of AI in the underwriting of life insurance policies.

Bibliography

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All writing must be in your own words. Any direct quotes from another author's work should be minimal and should be clearly shown in inverted commas (“”).

Various different citation styles are available. Unless you are told to use one particular style, the most important thing is be consistent throughout your document. For more help on citations and referencing, see Cite Them Right.

5.6 Memos/Emails

Memos are an older form of office communication in which a (typically) short message would be sent to a (typically) large number of employees. These have generally been replaced by email. If you do come across memos, the principles are similar to email.

5.6.1 Structure

Each company has its own style for preparing emails and memos. However, a typical structure and format for these forms of communication is outlined below.

- From: Your name/email address
- To: Recipient's name/email address
- Date: Date
- Subject: Subject of memo/email
- Introduction paragraph
- Main body with headings
- Conclusion/summary paragraph
- Regards
  Name

Subject
The subject of the email should summarise the purpose or content of the email. The wording of the subject gives the reader a first impression of the tone of the email. For example, if you were replying to a policyholder who is unhappy with the level of pension they are receiving, it might be better to use a more personal subject e.g., 'Your pension' rather than 'Annuity rates'.

Introductory paragraph
This is one paragraph which should only be a few sentences long. It should outline:

- The purpose of your email
- What your email is about

It might also make sense to reference previous communication with the recipient if relevant.

Summary/ conclusion paragraph
This section:

- Summarises the key points in your email.
- Does not introduce any new material.
- Is concise and friendly.
- Offers further help e.g., 'please give me a call if you have any questions'.

If the summary leads to unnecessary repetition, then it is best to leave this out - it best to use judgement in this case depending on the length of the full email.

5.6.2 Language and Tone

The language should be clear and simple to understand. If you are disagreeing with the recipient, be polite and try not to be dismissive e.g., do not say 'You are wrong'. If you are communicating 'bad news', try to start and finish the email with an upbeat tone if this is possible without seeming insincere.

When replying to complaints (which will inevitably happen from time-to-time), it is best to be clear, direct and (of course) respectful. It is generally unwise to try to convince the complainant that they should be happy with the situation and you should never try to avoid the issue. It is often sufficient to explain why the position is what it is, clear up any misunderstandings and explain any further options the complainant has.

5.7 Reports

Reports are usually the longest form of communication that actuaries can be asked to draft. As a result, they need to be structured carefully to ensure they ahave the desired impact.

5.7.1 Structure

Cover page

- This should usually include:
- Title of report
- Name of author(s)
- Name/Group for whom report is written
- Date of report

The title of your report should be concise but informative. For example, 'IFRS 17 Report 2024' is quite a vague title but 'Report on YE 2024 IFRS 17 results for Heriot-Watt Limited' is clear and concise.

Table of contents
A table of contents provides readers with a roadmap, allowing them to quickly locate specific information within the report. This can save time and help the reader efficiently navigate through the document. A well-structured table of contents demonstrates professionalism, enhances the overall presentation of the report and contributes to a positive impression of your work.

Executive summary (if included)
When structuring a report, a decision is needed on whether an executive summary is required. An executive summary presents the key information (including conclusions and/or recommendations) at the start of the report. These can be useful for individuals who don't need to know all of the details or who don't have time to read the full document. For example, you might be required to prepare a report for the board of directors of a company - the finance director might want to read the entire report while the managing director may only have time to read the summary.

If an executive summary is included, this should outline:

- Aim of report
- Sources of data
- Key points from main body of report
- Limitations of report
- Conclusions and recommendations

Introduction (if no executive summary is included) If an executive summary is not included, an introductory section is needed to set out:

- Aim of report
- Sources of data
- Limitations of report

Main body

Conclusion
Summaries key points and cover recommendations if relevant.

5.8 Presentations

When preparing a set of presentation slides, the slides should typically include the following structure:

- Title slide
- Introduction/ agenda slide
- Main body slides
- Summary/conclusion slide
- Questions slide
- References slide

Title slide:

- Title of presentation
- Presenter's name and job title
- Date

Introduction/Agenda slide:

- Provide an overview of the presentation's aims
- Introduce the topics that will be covered

Main body slides:

- Divide the main content into logical sections
- Keep text concise and to the point. It is important to limit the amount of text on each slide to avoid overwhelming the audience.
- All points in the 'General tips' section above remain relevant for the main body slides of a presentation(e.g., use of headings, bullet points, visual aids etc.)

Summary/ conclusion slide:

- Summarise the key points and ensure that you are emphasising the main takeaway points.
- Provide recommendations if relevant.

Questions/ discussion slide:

- Reserve a slide for audience questions or discussion points to encourage interaction/ engagement with the audience.

References slide:

- Include a slide listing your references or sources in the presentation.

5.9 Briefing/ Technical papers

These documents can be set out in a similar way to a report but should omit any sections that are not appropriate for your particular document. For example, you may not be required to make recommendations when preparing a briefing paper.

An executive summary may not be required for a briefing paper which is already quite short.

5.10 Articles

The advice under 'general structure' is relevant for preparing communication in the form of an article.

The format of an article should be similar to an email and should include:

- A title
- Introduction
- Headings
- Conclusion/Summary
- Author's name and date of article

5.11 Reviewing your communication

It is important to double check your work once you have prepared the communication. Once you are finished the initial draft, it is worth checking that:

- Each paragraph or idea is relevant to the reader
- Each paragraph helps to meet the aim of the communication
- There are no obvious key ideas missing from the communication
- The order of your ideas makes sense and is easy for the reader to understand
- The language used is simple without any jargon
- The tone is polite and neutral
- The content is factually accurate
- The grammar, spelling and punctuation has been checked.

Often a good question to ask yourself is "What would my manager say if I gave her this communication to review?"

5.12 Worked Example

5.12.1 Bad Communication Example:

Read the example below and consider how the communication could be improved:

To:j.murphy@watt.com
From:j.smnith@watt.com
Subject: Sales

Hey,

So I looked at the sales data, and things aren't looking great. Sales are down, customers are complaining, and the western region is a mess.

First, let's speed up deliveries. Customers hate waiting, and it's driving them away. Also, we should probably hire more people for customer support because people aren't happy. Finally, the western region is just doing bad. We should do something about that, maybe advertise more or something.

Let me know what you think. We should get this sorted soon.

Thanks,

John

Consider how the following elements of this email communication could be improved:

- Layout?
- Headings?
- Language?
- Tone?

5.12.2 Good Communication Example:

Now let's read of a better example of the same communication:

To:j.murphy@watt.com
From:j.smnith@watt.com
Subject: Analysis of Sales Data and Recommendations for Improvement

Hi Jane,

I hope this email finds you well.

I have completed an analysis of our recent sales data and have identified some areas where we can improve. I've summarised the key findings below and included a few recommendations to help boost our sales performance.

Key Findings:
The primary findings of my analysis of the sales data are as follows:

- Decline in Sales Growth: Over the past quarter, we've seen a 10 percent decrease in sales growth compared to the previous quarter.
- Customer Feedback: Customer feedback indicates dissatisfaction with product delivery times and customer support.
- Regional Disparities: Sales in the western region are 20 percent lower than in other regions, suggesting a need for targeted marketing.

Recommendations:
I believe that there are a number of actions that we can take to improve our sales. These include:

- Collaborate with the logistics team to reduce delivery times by implementing more efficient processes.
- Increase customer support staffing to ensure all customer inquiries are addressed within 24 hours.
- Develop a targeted marketing campaign to address the decline in sales in the western region.

Please let me know if you'd like more information on these findings or if you'd like to discuss these recommendations in detail. I look forward to working with you to implement these changes.

Kind regards,
John

5.12.3 Analysis:

In the "Good Communication" example, the email has a clear subject line and is structured with headings to guide the reader through the key findings and recommendations. It uses a professional tone and provides specific recommendations based on the analysis of the sales data.

In contrast, the "Bad Communication" example has a less descriptive subject line, a lack of structure, and a casual tone that might not be suitable for business communication. It doesn't clearly explain the findings, and the recommendations are vague and lack detail. The reader would likely have questions about what the specific issues are and what needs to be done to fix them.

5.13 Conclusion

In this chapter we have set out many different considerations for drafting communications. It is important to remember the various specific rules (e.g. what sections should be included in a report) as well as the overarching principles.

The next time you draft a communication, have a look at this chapter both before and after. Be your own ruthless editor when you rewrite. As time goes on, the principles from this chapter will become second nature to you.

Good communication is essential for actuaries, so never forget the golden rule: As a writer, your job is to help the reader understand the messages that you want to communicate. Everything should flow from this.